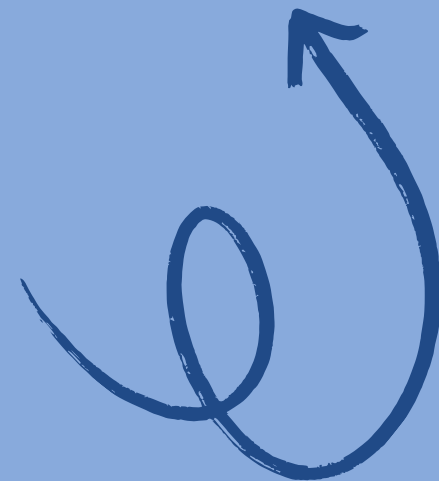


PathSense: Smart Cane Navigation

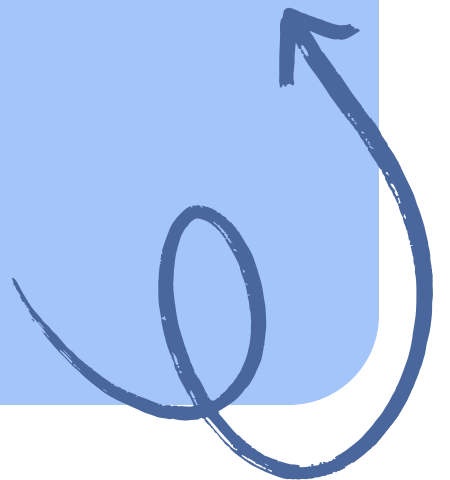
Team of 4 from Amal Jyothi
College



Limitations of Traditional Canes

Key Challenges for Visually Impaired Users

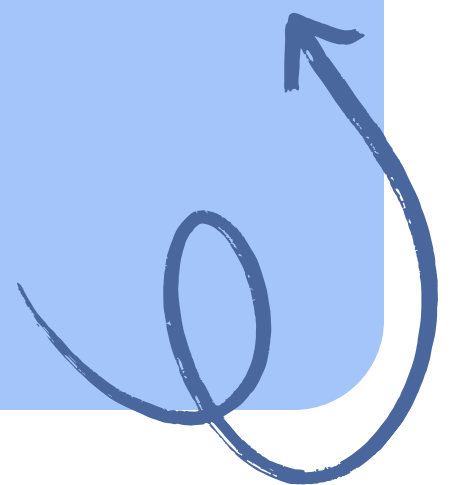
Traditional white canes primarily detect ground-level obstacles, **missing** hazards at chest and head height while lacking depth awareness for potholes and curbs, limiting user safety during navigation.



Why It Matters

Importance of Addressing Navigation Challenges

India has one of the **largest visually impaired populations**, especially in Kerala, where uneven roads pose significant risks. Our device addresses critical gaps in obstacle detection and emergency response.



Our Solution: Enhanced Navigation Safety



01

Side Detection

Dual sensors identify obstacles on both sides.

02

Depth Sensing

Sensor at the tip detects height changes.

03

Emergency Alert

SOS button sends alerts during emergencies.

System Flow Overview

Sensing

01

Sensors continuously scan the path ahead and below.

Processing

03

A microcontroller checks sensor values against thresholds.

Alerting

02

Vibration feedback guides the user's next action.

Integration

04

Ultrasonic and depth sensors work in harmony.

What Makes PathSense Unique



01

Coverage

Detects forward, side, and ground depth obstacles.

02

Emergency

Integrated SOS alert for fall response included.

03

Precision

Dedicated sensors specifically target potholes and puddles.

Component Placement on the PathSense Cane



Handle

Microcontroller and SOS button housed neatly.



Grip

Vibration motor for immediate silent feedback.



Sensors

Ultrasonic and depth sensors ensure navigation safety.

Practical Strengths of PathSense



01 Low Cost

Estimated build cost is **affordable** for users.

02 Proven Components

All core parts are **well-documented** and accessible.

03 Retrofit Design

Attaches to standard canes, requiring **no custom builds**.

Next Steps

Sensor Logic



Finalizing and testing sensor logic in simulation ensures the functionality works as expected before moving to the physical prototype, minimizing potential issues during the build phase.

Component Sourcing



Sourcing components for the physical prototype will begin once the simulation is validated, allowing us to gather all necessary parts for assembly efficiently and effectively.

Field Testing



Conducting field tests on real pavement conditions will help us refine the sensor thresholds and ensure accurate obstacle detection and responses in varied environments.

Feature Exploration



Exploring SMS and GPS integration for the SOS feature will enhance user safety, providing reliable emergency communication options directly from the PathSense device during critical situations.

Thank You



PathSense: Confident Navigation for All